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2025
2026





Hardware–Software Co-Design of a Quantized HiFiC GAN Accelerator for Learned Image Compression on FPGA

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1. Problem statement: The Image Data Explosion

Massive Visual Data Growth

- **Surveillance:** 24/7 high-res monitoring
- **Medical:** Ultra-HD clinical imaging
- **Remote sensing:** High-fidelity GIS data
- **Social Media:** Global multimedia sharing

"Data generation volumes are outpacing the capacity of modern storage infrastructure."

- **181 zettabytes** of data were generated in 2025
- Videos account for ~82% of internet data traffic

source: <https://explodingtopics.com/blog/data-generated-per-day>



- **Rising Storage Costs**

HDDs and SSDs struggle to keep pace with petabyte-scale generation

- **Edge Processing Demand**

Critical need for real-time high-res compression on embedded IoT systems.

- **Legacy Limitations**

JPEG/JPEG2000 show artifacts at high compression ratios.

2. Research Objectives: HiFiC GAN Architecture

Table 2.1: Technical specifications of current studies

Tiêu chí	VVC/H.266 6 [1]	Cheng 2020 [2]	STF [3]	TCM [4]	HiFiC (SW) [5]
Year ↑	2020	2020	2022	2023	2020
Type	Traditional	CNN (VAE)	Transformer	CNN + Transformer	GAN
Backbone	H.266 standard	ResBlock + Attn	Swin Transformer	TCM Block	Conv + ResBlock
Optimized for	Rate- Distortion	Rate- Distortion	Rate- Distortion	Rate- Distortion	Perceptual
PSNR (dB, ~0.4 bpp) ↑	~32.0	~33.0	~33.8	~33.5	~28.5
MS-SSIM ↑	~0.94	~0.96	~0.97	~0.97	~0.91
LPIPS ↓	~0.40	~0.30	~0.25	~0.23	~0.15
BD-Rate vs VVC ↓	0% (ref)	~-2%	~-8%	~-10%	N/A
Model Params ↓	N/A	~115M	~599M	~50M	~200M
Memory (FP32) ↓	—	~440 MB	~2.3 GB	~190 MB	~800 MB

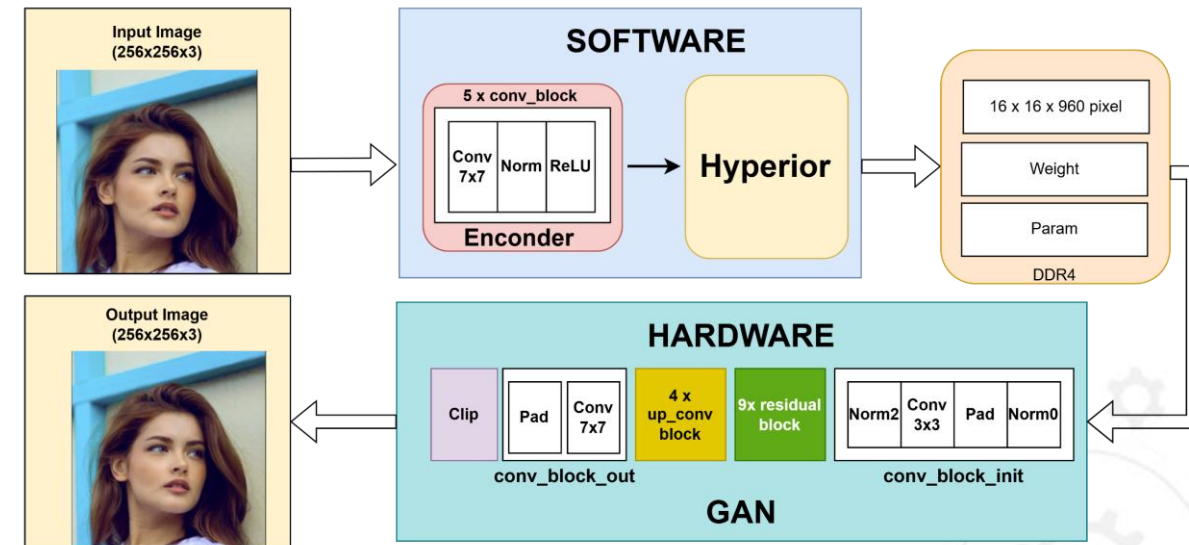


Figure 2.1: System-level hardware-software co-design and design flow

2. Research Objectives

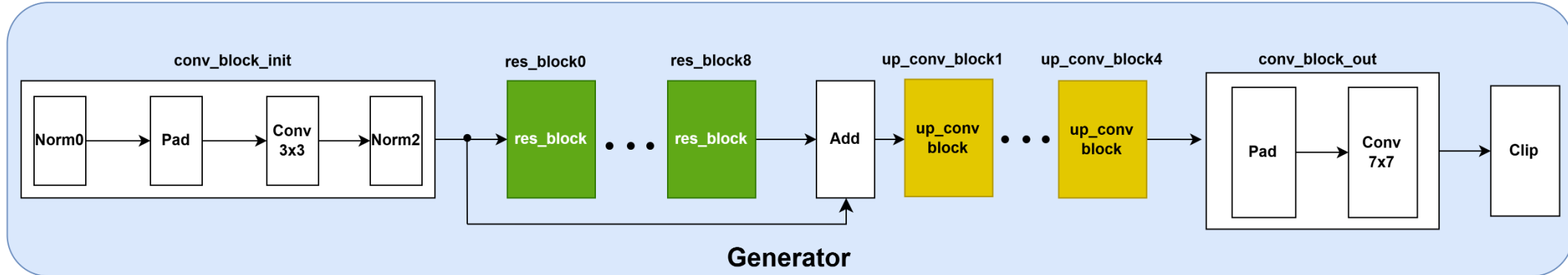


Figure 2.2: GAN Generator architecture

- **The Data:** CPU runtime profiling (Intel Core i9 @ 2.2 GHz) reveals a severe imbalance in execution time.
- **The Engineering Opportunity:** Every ResBlock shares identical computational configurations and tensor sizing (Input: 18x18x960, Output: 16x16x960). By designing a single, highly optimized hardware accelerator, we can reuse it across all 9 ResBlocks.

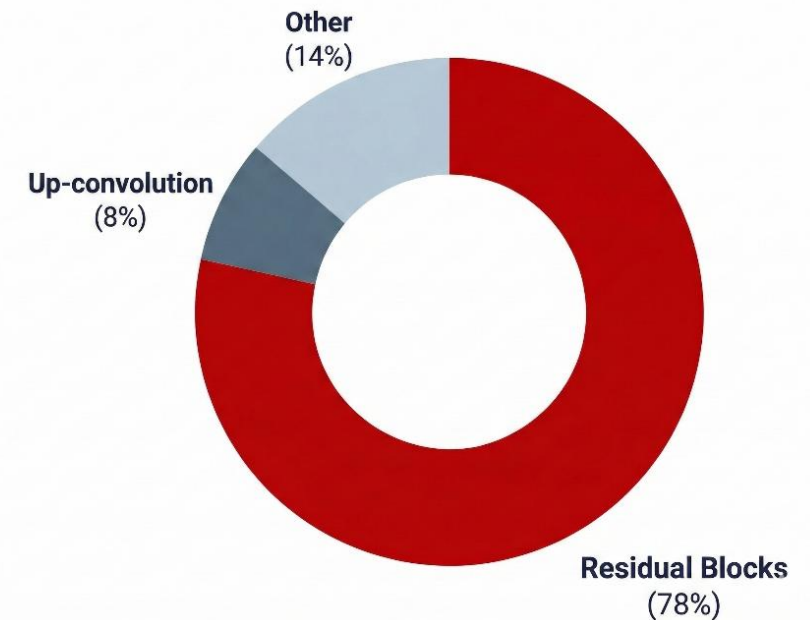


Figure 2.3: Execution time distribution of GAN Generator blocks

2. Research Objectives

Hardware–Software Partitioning Strategy

- **Hardware (PL — GAN Accelerator):** Offload all 9 Residual Blocks to custom FPGA logic → eliminates the 78% bottleneck
- **Software (PS — ARM Cortex-A53):** Encoder, Hyperprior, Up-convolution blocks, and system control remain in software

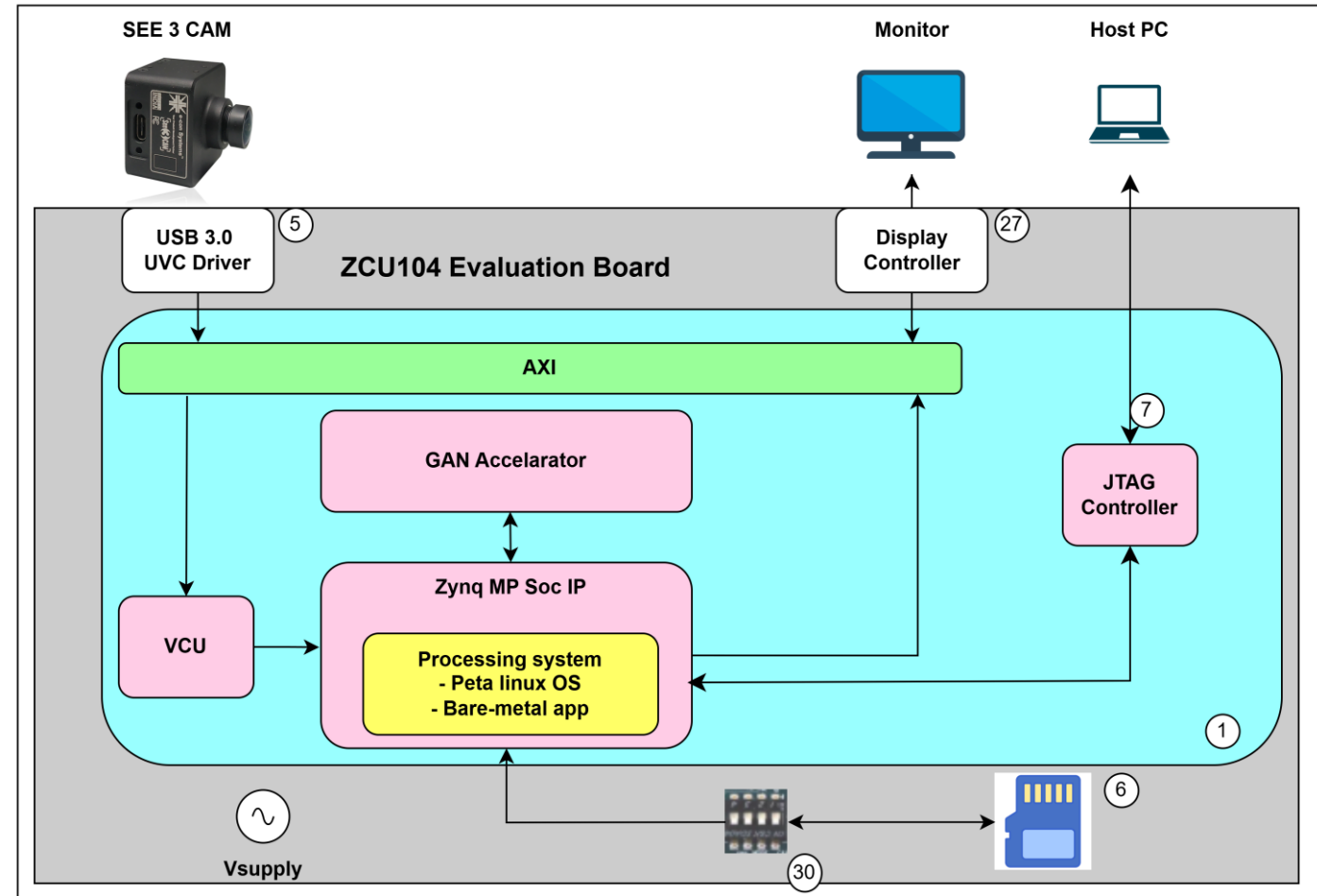


Figure 2.3: System Architecture

2. Research Objectives

Design Specifications

ARCHITECTURE & PLATFORM:

- Core Design: **Residual Block (Resblock) IP for HiFiC**
- Target Device: Xilinx Zynq UltraScale+ MPSoC ZCU104
- Development Flow: Vitis HLS & Vivado
- Operating Frequency: 200 MHz

Performance & Real-Time Execution

- Input Resolution: $16 \times 16 \times 960$
- Throughput: ≥ 30 GOPS
- Quantization: Mixed-precision (FP16 Mixed, FP32 Precision Base)

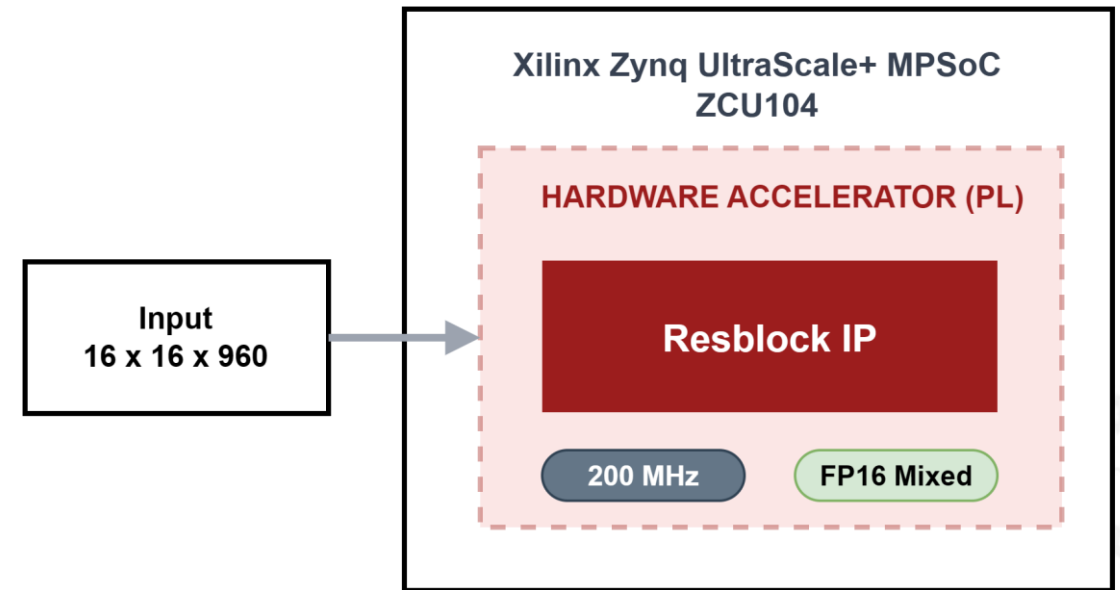
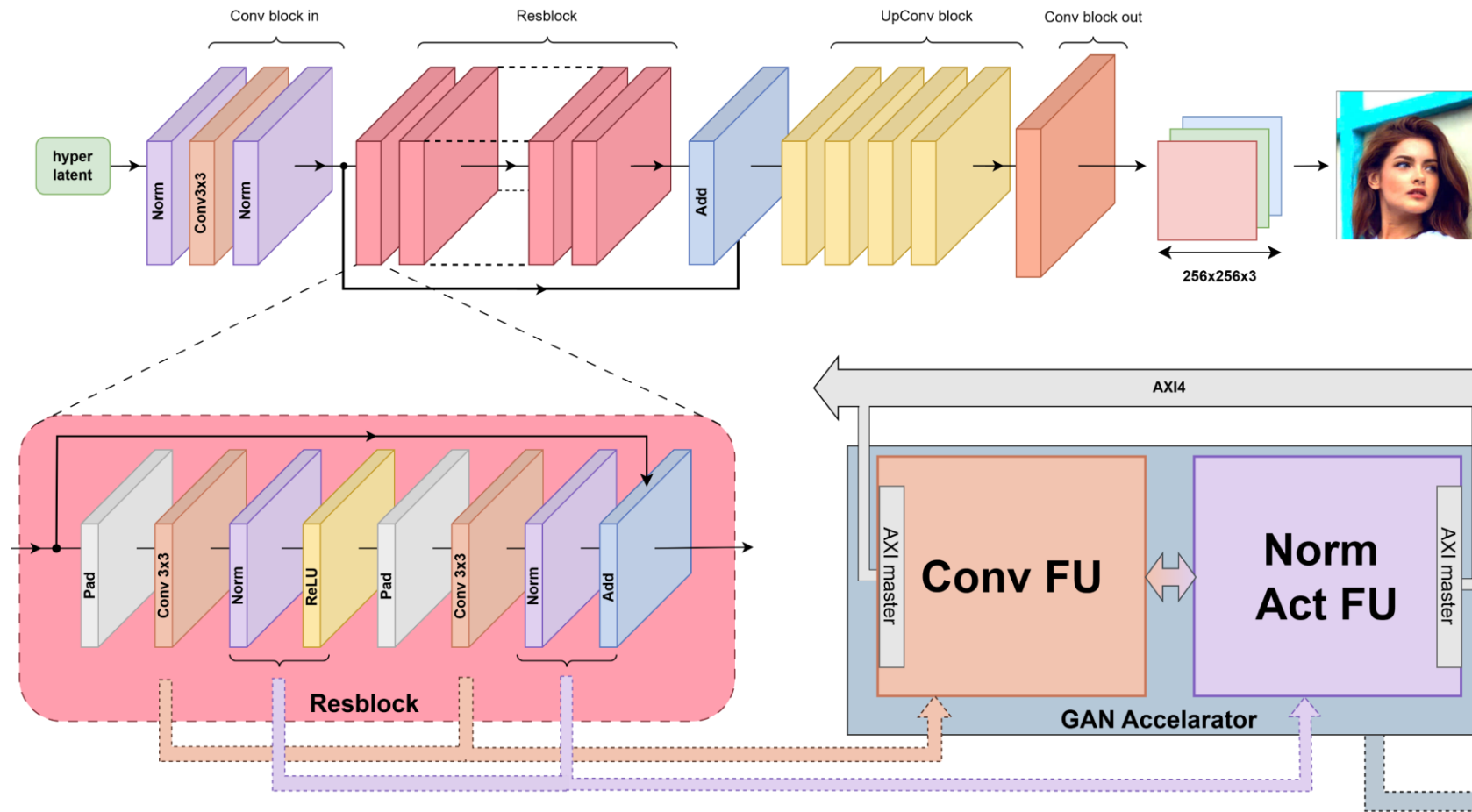


Figure 2.6: Specification of resblock architecture

2. Research Objectives



→ Design a computational accelerator suitable for the HiFiC network, consisting of two main processing units: **Conv FU** and **Norm Act FU**.

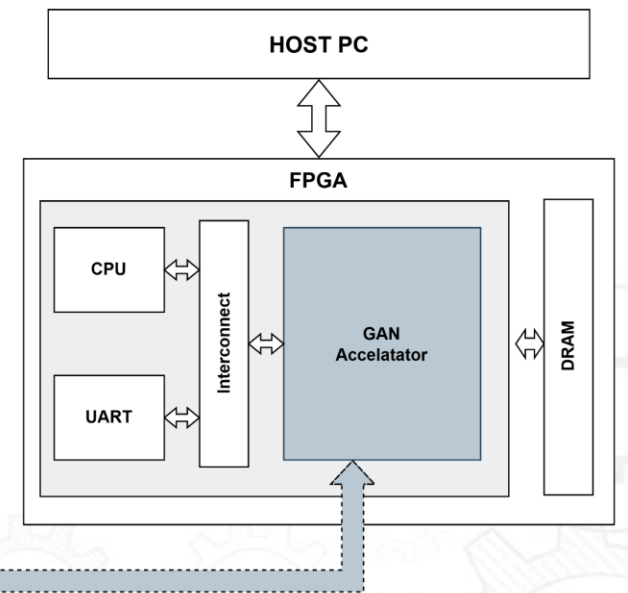


Figure 2.5: Overview about system and resblock architecture

2. Research Objectives

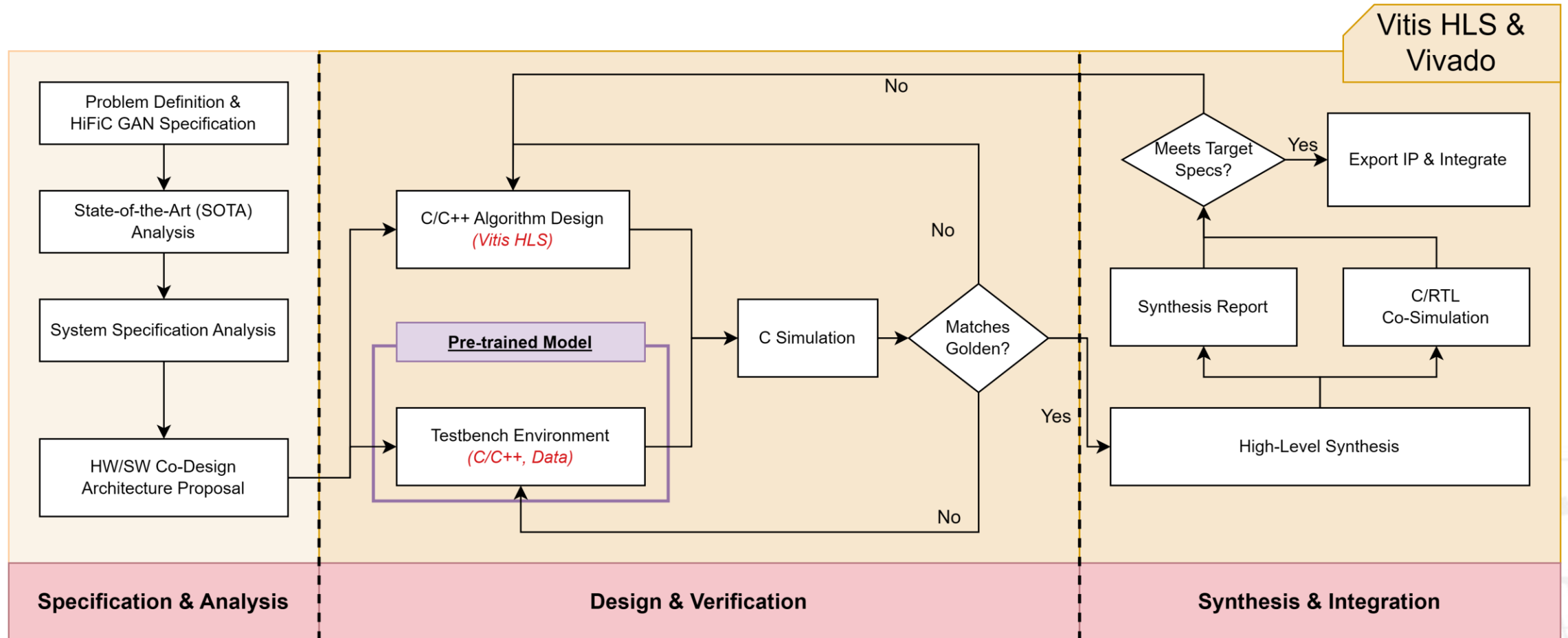
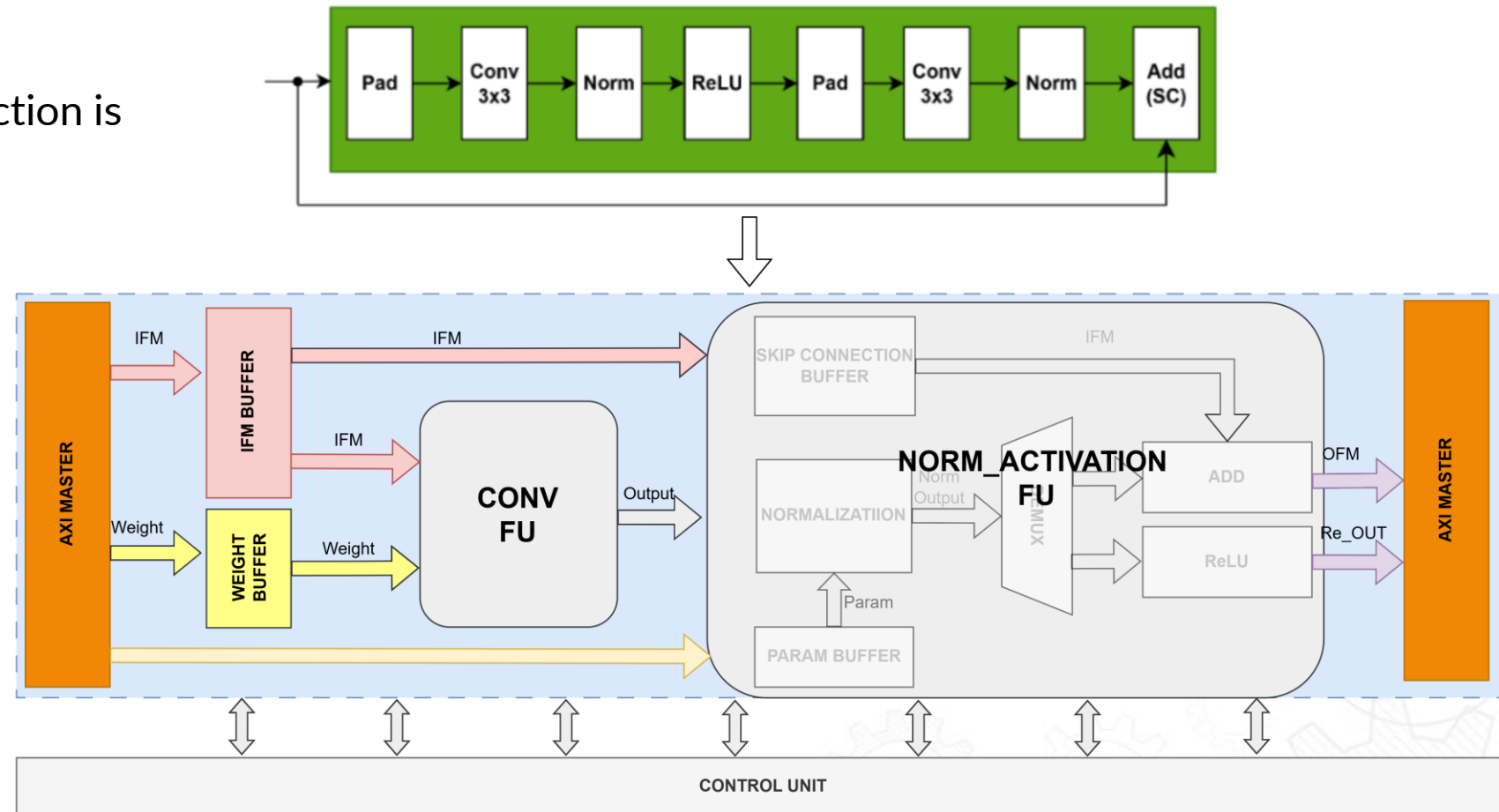


Figure 2.7: Project workflow

3. Proposed Architecture

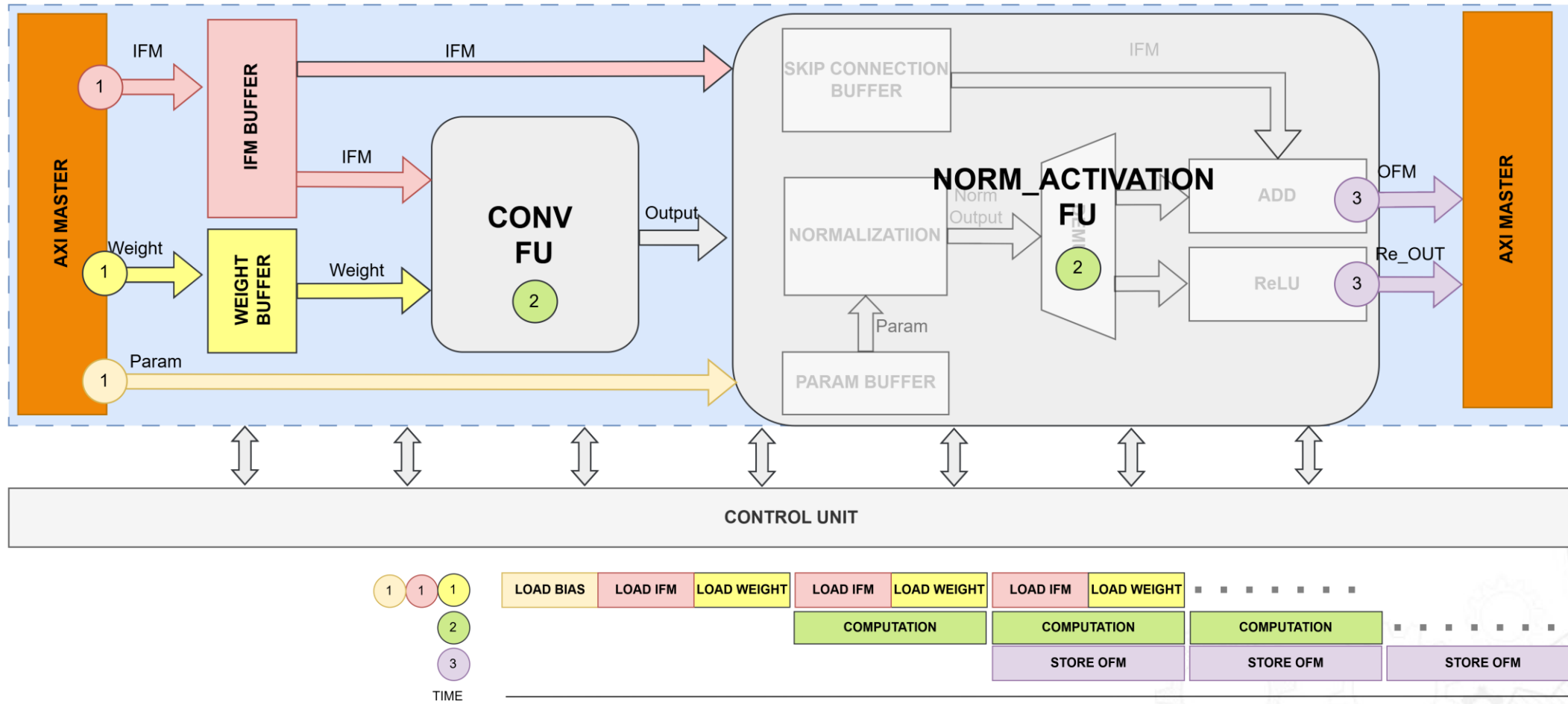
- Resblock accelerator architecture:

- Convolution is calculated in CONV FU
- Channel Norm, activation function is calculated in in NORM_ACTIVATION FU



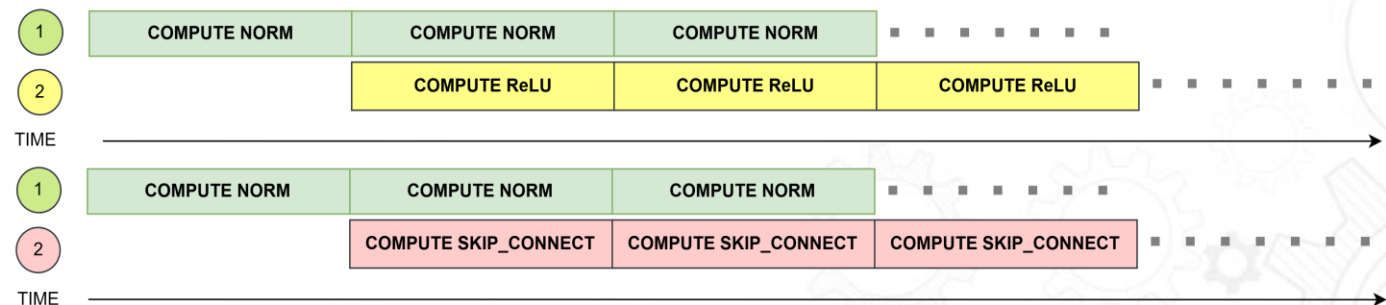
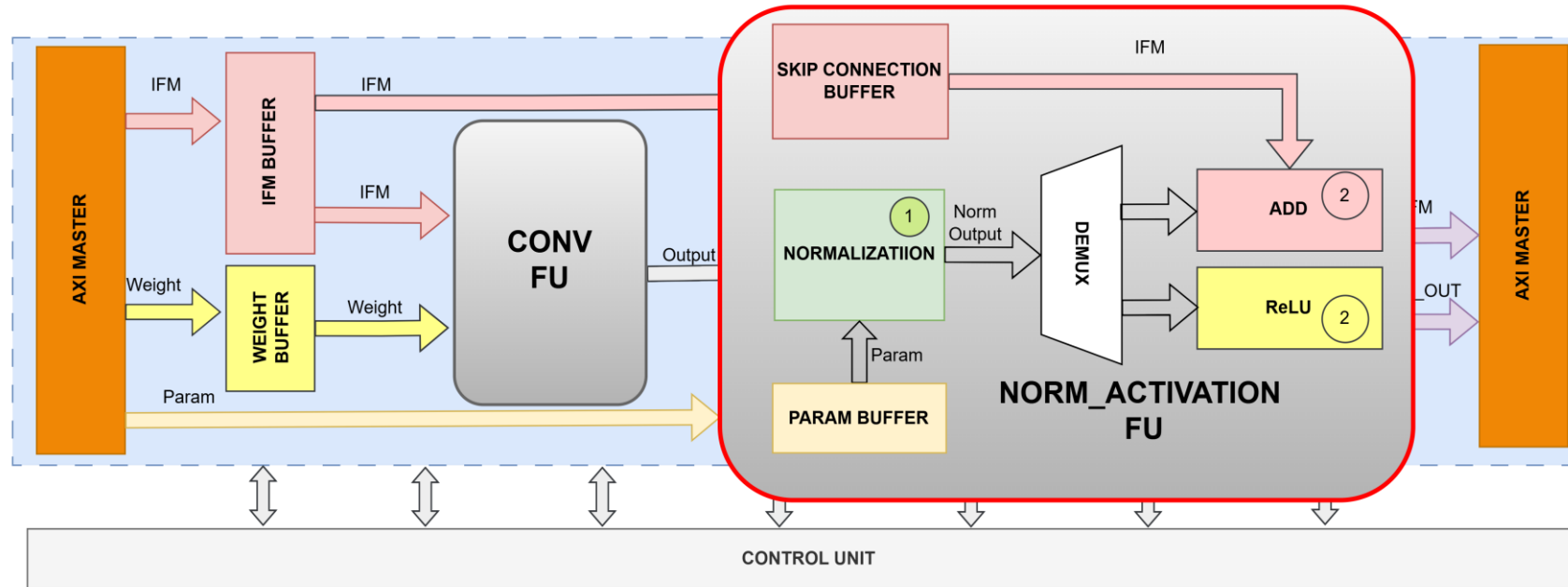
3. Proposed Architecture

Dataflow of accelerator:



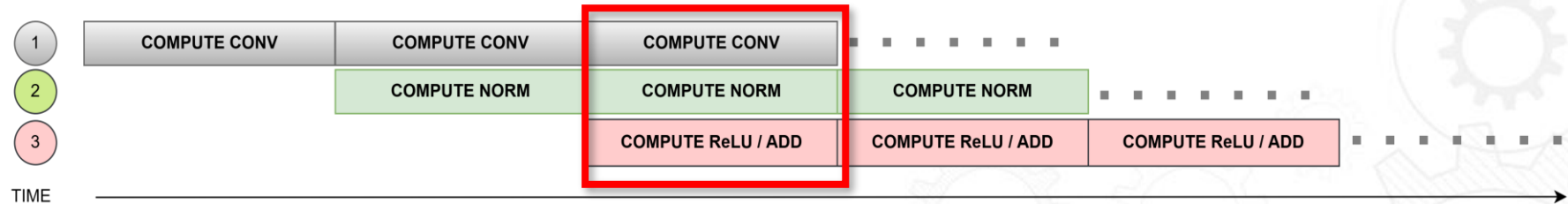
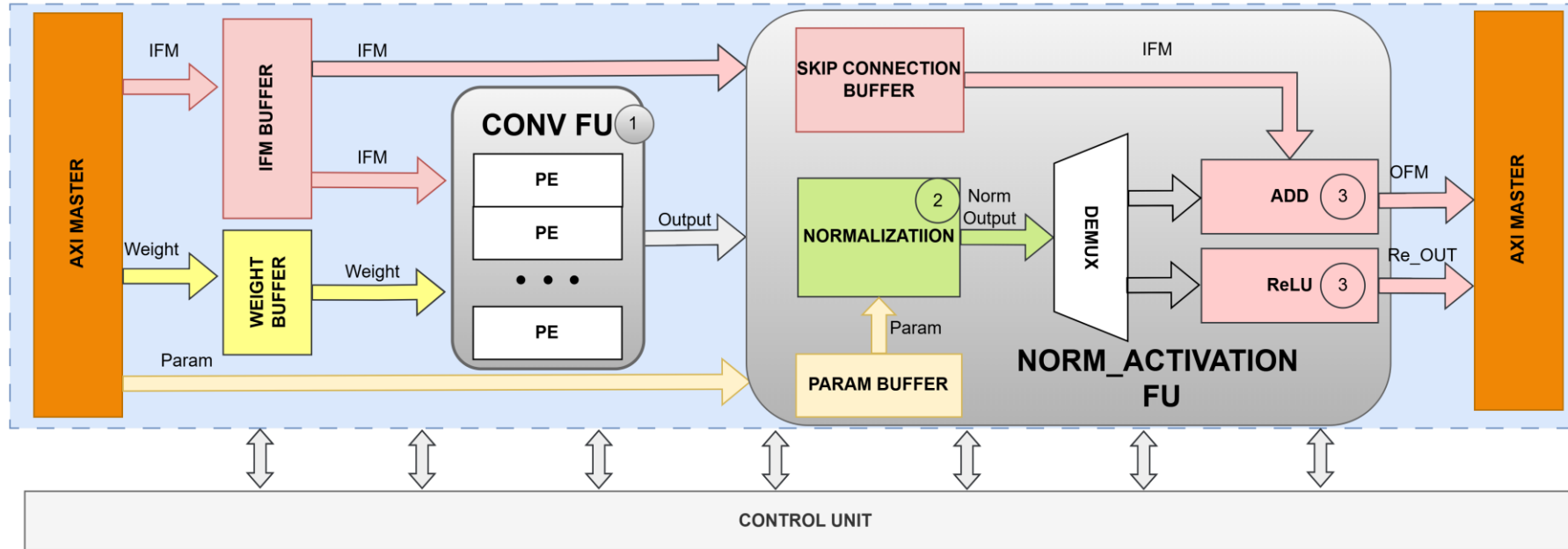
3. Proposed Architecture

C_NORM_ACTIVATION Functional Unit Behavior



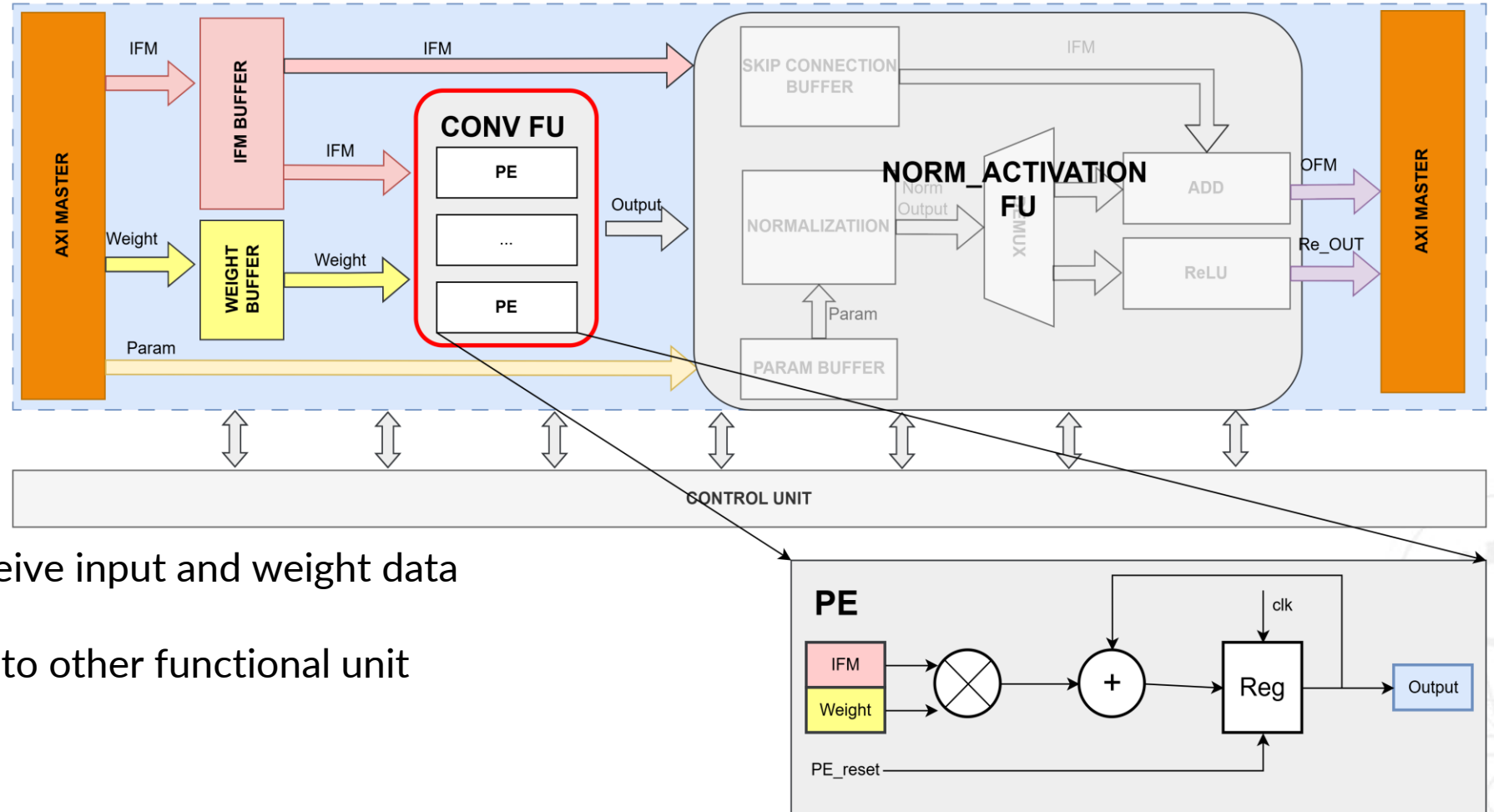
3. Proposed Architecture

Using pipeline technique to calculate 3 layers



3. Proposed Architecture

- Processing Element (PE) structure:



- Processing elements receive input and weight data from local buffers
- OFM data is transferred to other functional unit

4. Testing Environment

Testing Environment and Accuracy Evaluation

Objectives of the testing environment:

- Verify the functional correctness of the HiFiC GAN accelerator on ZCU104 FPGA
- Measure key performance metrics of the hardware-software co-design system

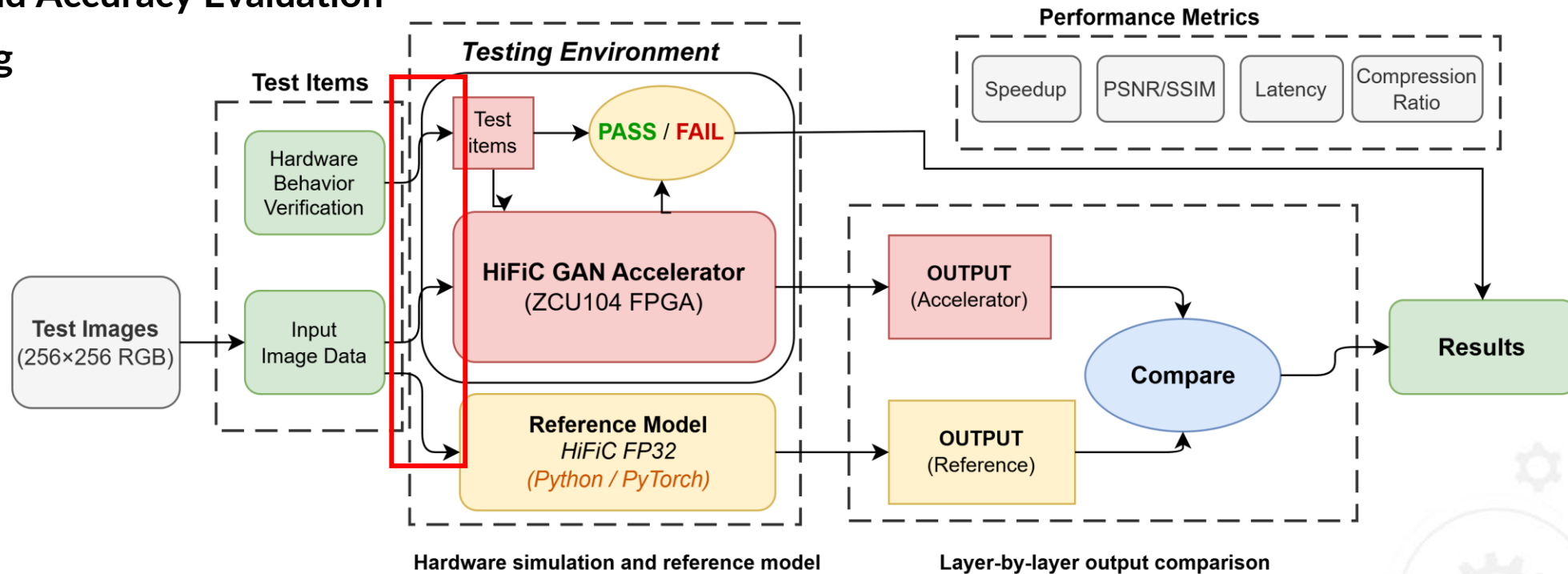


Figure 4.1: Verification flow and performance measurement environment

Input Data:

- Test vectors for hardware behavior verification of the ResBlock accelerator
- Random 256x256 RGB images used as compression inputs (sampled from standard image datasets)

4. Testing Environment

Testing Environment and Accuracy Evaluation

Verification process:

- Compare reconstructed image quality (PSNR, SSIM) between the FPGA accelerator output and the FP32 reference model
- Validate bitstream size and compression ratio consistency between FP16 hardware and FP32 software

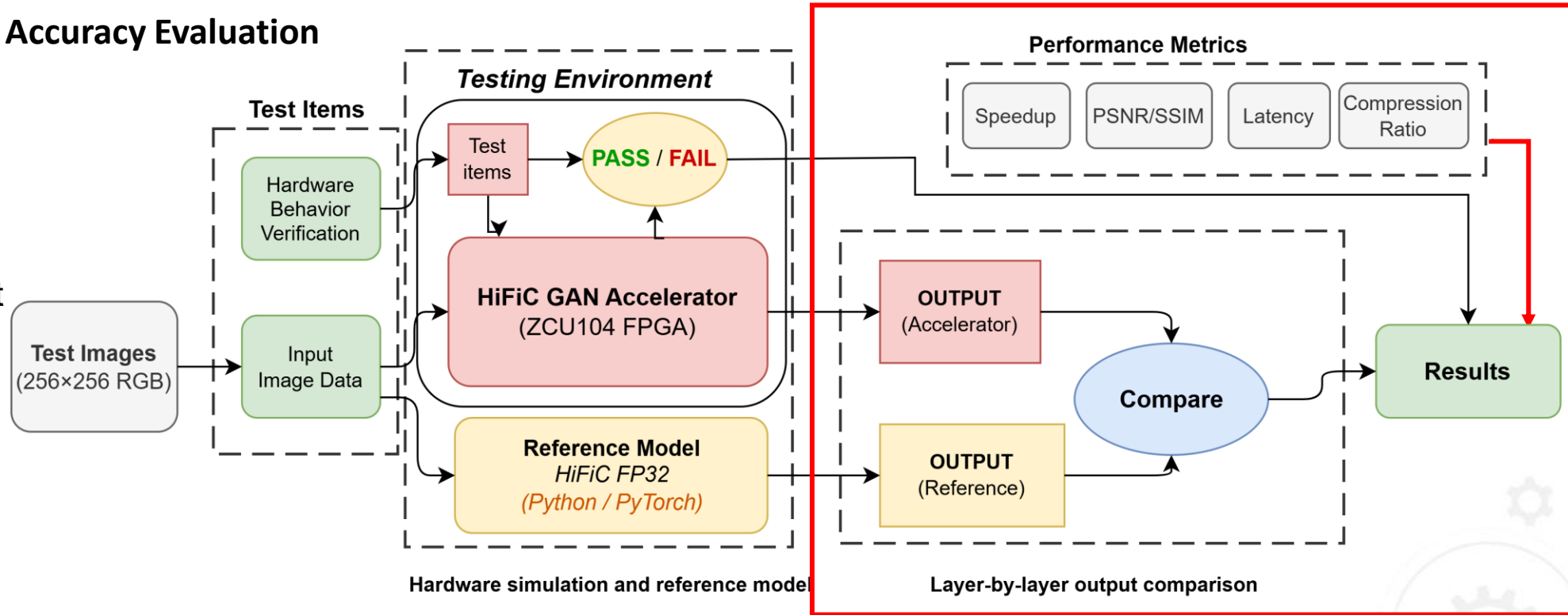


Figure 4.1: Verification flow and performance measurement environment

Accuracy evaluation:

- Computational accuracy of the FP16 hardware is evaluated against the FP32 HiFiC reference model (Python / PyTorch)
- Accepted threshold: PSNR degradation < 0.03 dB, SSIM delta < 0.0001

System Integration on ZCU104



Control: PetaLinux OS

Bus: AXI4-Master Interface

Output: Display Controller

The diagram illustrates the system architecture of a ZYNQ UltraScale+ MPSoC. The central component is the **ZYNQ UltraScale+ MPSoC**, which is connected to several peripheral blocks:

- Processor System Reset**: This block provides reset signals to the system, including `slowest_sync_clk`, `ext_reset_in`, `aux_reset_in`, `mb_debug_sys_rst`, and `dcm_locked`. It also outputs `mb_reset`, `bus_struct_reset[0:0]`, `peripheral_reset[0:0]`, `interconnect_aresetn[0:0]`, and `peripheral_aresetn[0:0]`.
- smartconnect_1**: An **AXI SmartConnect** block that manages the interconnect between the processor and the system. It receives `S00_AXI`, `S01_AXI`, and `S02_AXI` signals and outputs `M00_AXI`, `M01_AXI`, and `M02_AXI`.
- smartconnect_0**: Another **AXI SmartConnect** block that manages the interconnect between the processor and the system. It receives `S00_AXI` and outputs `M00_AXI`.
- system_ila_0** and **system_ila_1**: **System ILA** blocks used for logic analysis. `system_ila_0` is connected to `SLOT_0_AXI`, `SLOT_1_AXI`, `SLOT_2_AXI`, and `SLOT_3_AXI`. `system_ila_1` is connected to `SLOT_0_AXI`.
- resblock_top_0**: A **Resblock_top (Pre-Production)** block that provides `s_axi_control` and `ap_clk` signals to the processor. It also outputs `ap_rst_n` and `interrupt`.
- rst_ps8_0_96M**: A **Processor System Reset** block that provides reset signals to the system, including `slowest_sync_clk`, `ext_reset_in`, `aux_reset_in`, `mb_debug_sys_rst`, and `dcm_locked`. It also outputs `mb_reset`, `bus_struct_reset[0:0]`, `peripheral_reset[0:0]`, `interconnect_aresetn[0:0]`, and `peripheral_aresetn[0:0]`.

The ZYNQ UltraScale+ MPSoC is configured with various parameters, including `pl_resetn0`, `pl_clk0`, `pl_clk1`, and `pl_ps_irq0[0:0]`. It also features a `maxihpm0_fpd_acik` and `saxihp0_fpd_acik` signal.



5. Result

Resource breakdown for the ResBlock accelerator on Zynq UltraScale+ ZCU104

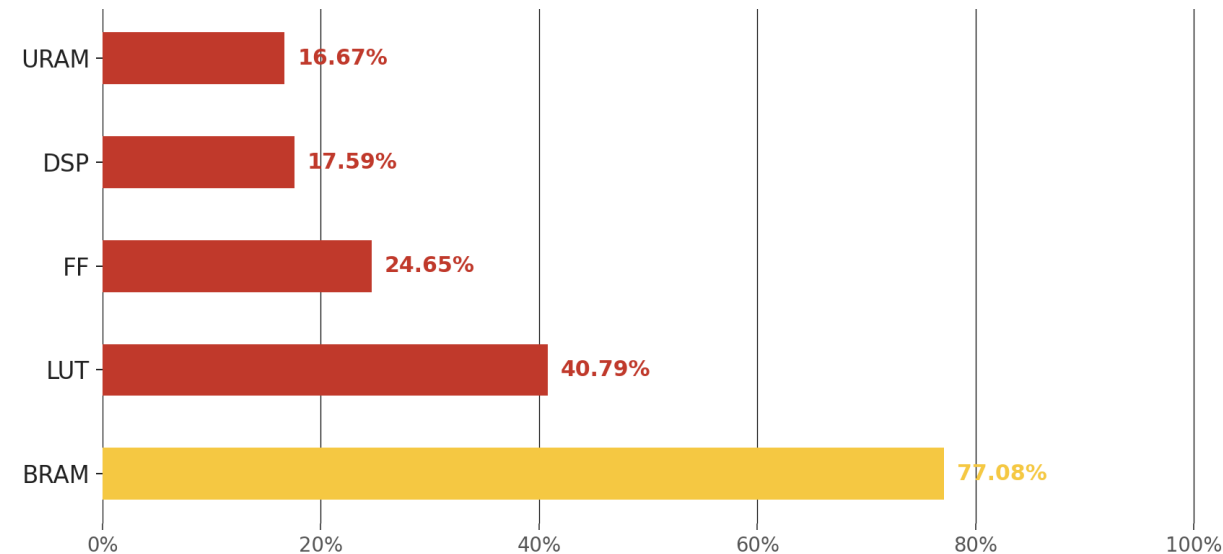


Figure 4.3: Resource Utilization

☰ Detailed Resource Count

RESOURCE TYPE	USED	AVAILABLE	UTILIZATION
 URAM	16	96	16.67%
 DSPs	304	1,728	17.59%
 FlipFlop	113,575	460,800	24.65%
 LUTs	77,324	230,400	40.79%
 BRAM	240.5	312	77.08%

5. Result

Performance Metrics and Speedup

Operating Frequency

250 MHz

Target (200 MHz) Exceeded

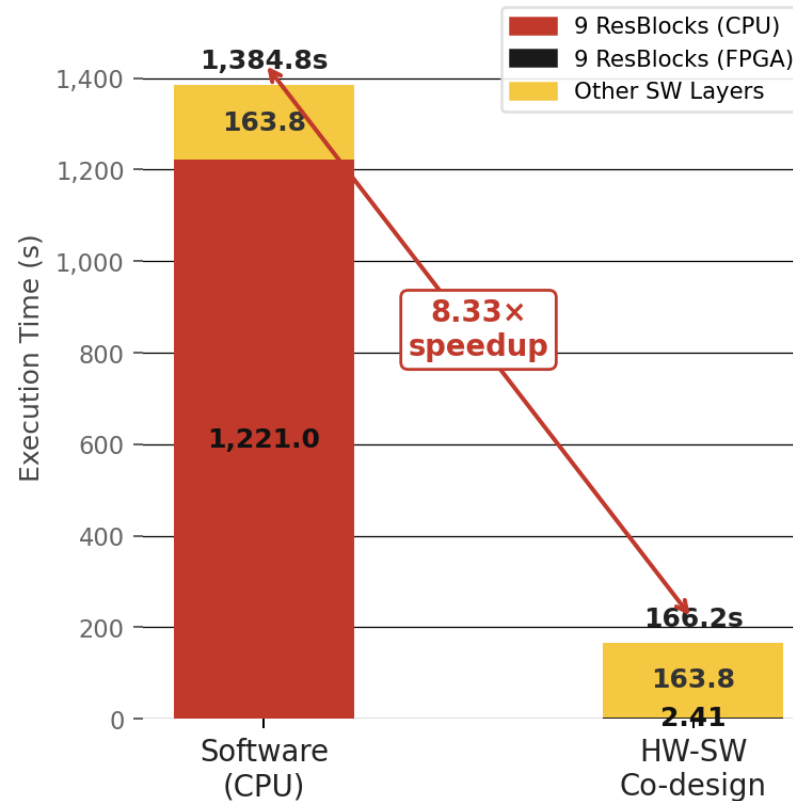
GOPS

31.7

KEY ACHIEVEMENT

Sustained steady-state throughput achieved via pipelined spatial parallelism and optimized memory hierarchy.

End-to-End Execution Time



Single ResBlock: FPGA vs Core i9

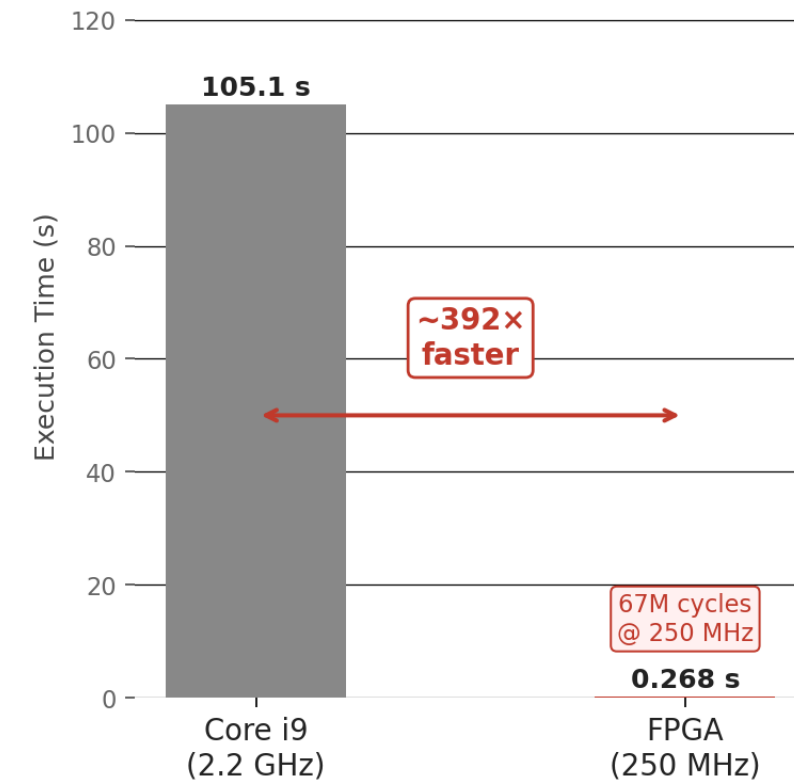


Figure 4.4: Execution Comparison

5. Result

Validation of Mixed-Precision Quantization for FPGA Deployment



3.97x
COMPRESSION RATIO
Maintained with FP16



74.82%
SPACE SAVING
+0.02% Improvement



0.03 dB
PSNR DEGRADATION

Performance Comparison

Metric	FP32	FP16 Mixed
PSNR (dB)	19.08	19.05
SSIM	0.8315	0.8314
Bitstream (B)	6,522	6,517
Bits Per Pixel	0.7961	0.7955

Visual Consistency Analysis

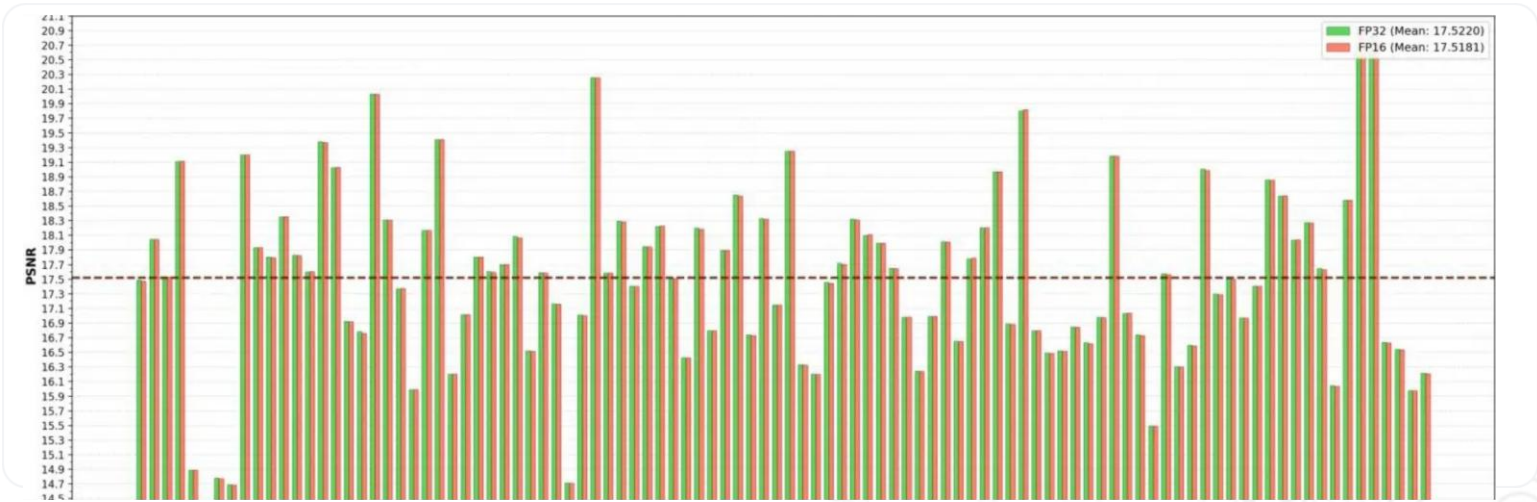


Figure 4.5: PSNR comparison across 100 images

6. Conclusion and future development

- **Conclusion:**

1. System exceeds initial design specification with operating frequency 250 Mhz.
2. Accelerator achieves 31.7 GOPS.
3. Power: 9.238W
4. Maintain the image quality compare with origin model
5. 392x faster than CPU

- **Future development:**

1. Implement to accelerate the whole GAN architecture on FPGA.
2. Optimize hardware resource utilization.
3. Implement multicore accelerator for video compression.

Achievement





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Thank you for your attention

References

- [1] B. Bross et al., "Overview of the Versatile Video Coding (VVC) Standard," *IEEE TCSVT*, vol. 31, no. 10, pp. 3736–3764, 2021.
- [2] Z. Cheng et al., "Learned Image Compression with Discretized Gaussian Mixture Likelihoods and Attention Modules," *CVPR*, pp. 7939–7948, 2020.
- [3] R. Zou et al., "The Devil Is in the Details: Window-Based Attention for Image Compression," *CVPR*, pp. 17492–17501, 2022.
- [4] J. Liu et al., "Learned Image Compression with Mixed Transformer-CNN Architectures," *CVPR*, pp. 14388–14397, 2023.
- [5] F. Mentzer et al., "High-Fidelity Generative Image Compression," *NeurIPS*, vol. 33, pp. 11913–11924, 2020.